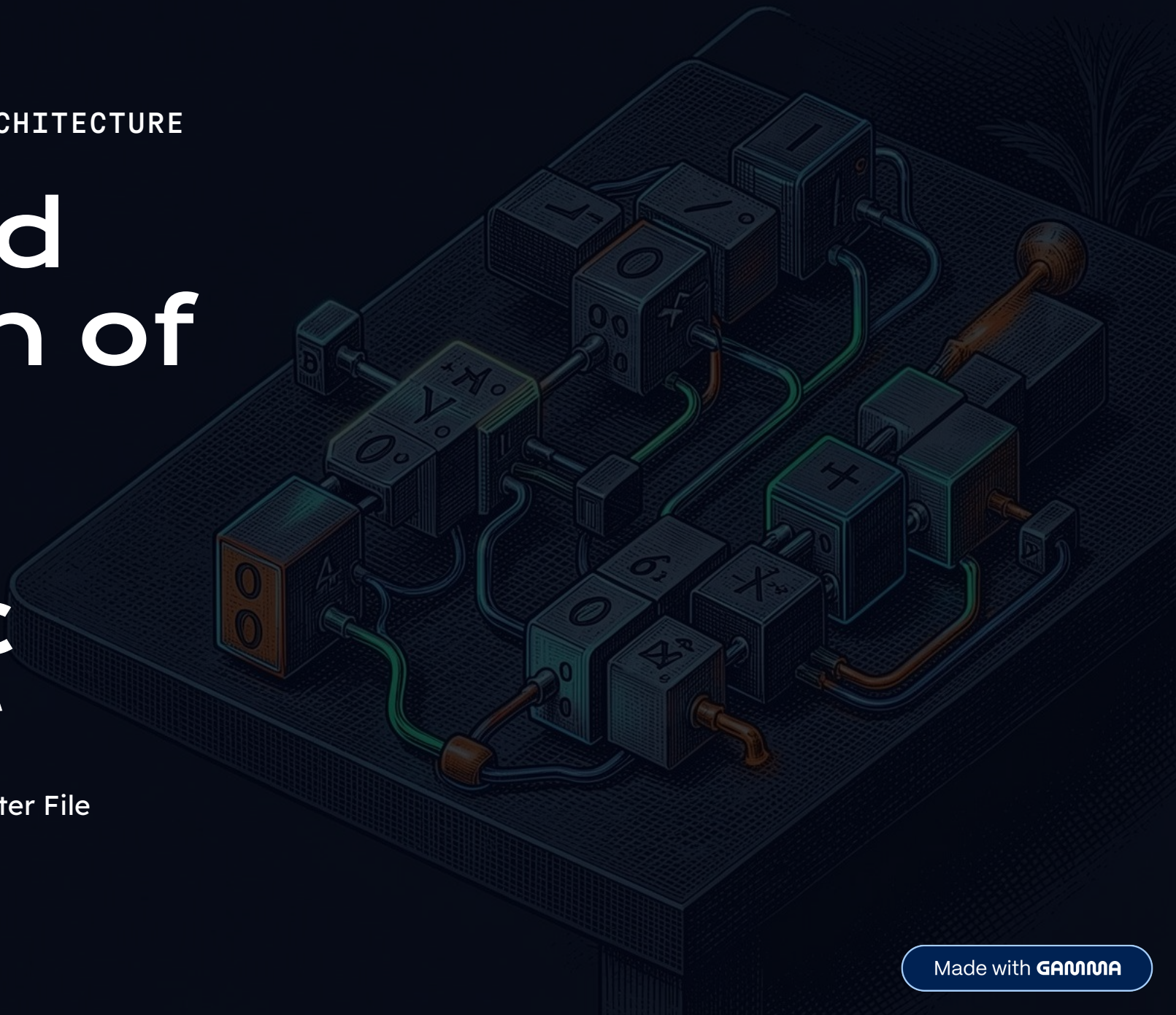
DIGITAL DESIGN AND COMPUTER ARCHITECTURE (DDCA)

Design and Simulation of a Four-Bit Binary Arithmetic Calculator

Module 4: Registers, Memory and the Register File



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SYSTEM PURPOSE

Binary inputs become a verified arithmetic result

Binary arithmetic uses combinational logic to transform current inputs into outputs.

```
INPUT → LOGIC → RESULT
```



Four-bit operands

Accepts two binary operands.



Arithmetic output

Produces a four-bit sum plus a possible carry-out.



Verification

Design, implement, simulate in Logisim Evolution, and verify expected outputs.



Module 4 link

Registers store operands and results while the arithmetic circuit performs the calculation.

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CIRCUIT ARCHITECTURE

Four full adders form the ripple-carry path

Four-bit values range from 0000 to 1111. Combinational outputs depend only on the current inputs.

AND, OR, XOR and NOT gates build half adders and full adders.

FULL-ADDER EQUATIONS

Sum = A XOR B XOR Cin

Carry = (A AND B) OR (Cin AND (A XOR B))



Illustrative labelled diagram: four full-adder stages from A0–A3 and B0–B3, carry chain C0–C4, outputs S0–S3, and final Cout.

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ILLUSTRATIVE CIRCUIT

Logisim makes the datapath visible

Input switches

Feed logic gates or full-adder blocks.

Carry signals

Move stage by stage to the final result.

Output LEDs

Show S0–S3 and Cout.

CPU datapath: registers hold operands → arithmetic circuit processes them → result returns to a register.

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VERIFICATION PLAN

Expected cases define what to verify

A	B	Expected result
0011	0010	0101
0111	0001	1000
1111	0001	1 0000

Full-adder truth table: inputs A, B, Cin; outputs Sum and Cout.

OVERFLOW CHECK

Cout preserves the fifth bit

Cout preserves the fifth bit when the four-bit sum overflows.

[Insert Logisim simulation screenshot 1]

[Insert Logisim simulation screenshot 2]

PENDING EVIDENCE

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ARCHITECTURE CONNECTION

One small adder demonstrates a CPU building block

PROJECT AND MODULE 4

The project demonstrates binary addition with combinational logic and a visible carry chain. Applications include ALUs, processors, calculators and digital systems. Registers temporarily store operands/results; the register file provides fast CPU working storage; memory holds instructions and data.

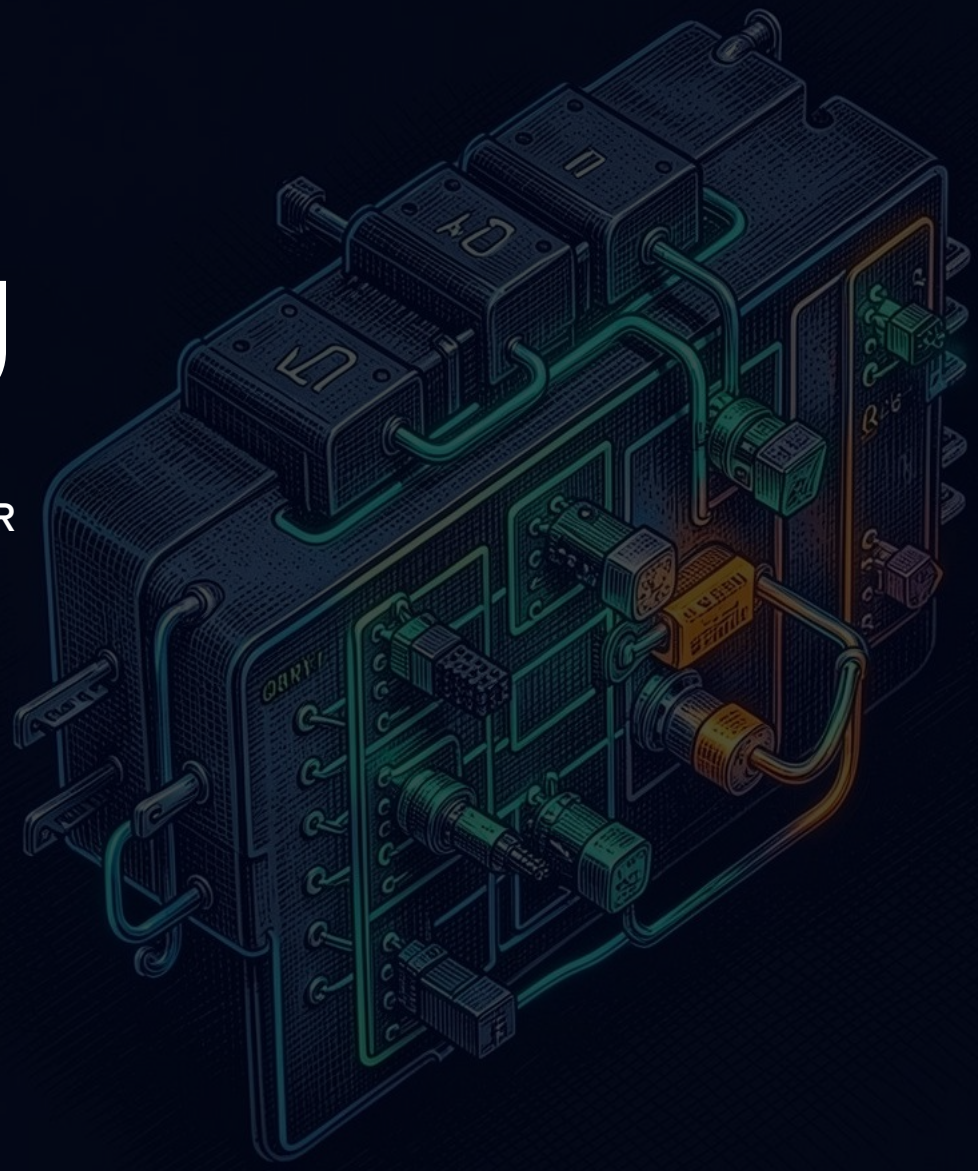
PERFORMANCE AND NEXT STEPS

Memory hierarchy and data-transfer delays can limit CPU performance. Implemented: four-bit addition logic and expected-result cases. Proposed: subtraction, more operations, seven-segment display, operand registers and a simple CPU datapath.

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THANK YOU

BINARY LOGIC · COMBINATIONAL DESIGN · COMPUTER ARCHITECTURE



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